Cloud IT Online

Cloud IT Online Batch-14

DATA STRUCTURE
&
Algorithm

সরকারি চাকরির পরীক্ষাভিত্তিক সাজেশন

◆ বিগত প্রশ্নের বিশ্লেষণ ◆ Topic-wise Syllabus ◆ সম্ভাব্য প্রশ্ন

Target Exams

◆ BB AD (ICT) ◆ BCS/PSC ◆ Assistant Programmer ◆ SO (IT) ◆ Bank & Other ICT Recruitment Exams

[এই গাইডটি Cloud IT Online Batch-এ ২০১৬–২০২৬ সাল পর্যন্ত অনুষ্ঠিত বিভিন্ন সরকারি ও ব্যাংক নিয়োগ পরীক্ষায় আসা Data Structure বিষয়ক প্রশ্নের ধরন অনুসরণ করে তৈরি করা হয়েছে। এতে সম্ভাব্য পরীক্ষার প্রস্তুতির জন্য সাজেশনমূলক প্রশ্ন অন্তর্ভুক্ত করা হয়েছে। প্রশ্নগুলো ভালোভাবে অনুশীলন করলে এই অধ্যায়ের জন্য একটি সুন্দর, গোছানো ও কার্যকর প্রস্তুতি সম্পন্ন হবে।]

1. Priority-wise Topic Analysis

Priority	Topic	Study
Priority 1 — High	Stack, Queue, Linked List, Tree, BST, Sorting, Searching, Time Complexity	Definition, difference, algorithm, pseudocode, complexity, hand-worked examples
Priority 2 — High	Infix/Postfix, Heap, Tree Traversal, BFS/DFS	Expression evaluation, tree/heap construction, traversal order, code writing
Priority 3 — High	Divide & Conquer, Dynamic Programming, ADT, Recursion, Hashing, Greedy Algorithm	Conceptual comparison, small examples, complexity analysis

2. Topic-wise Complete Suggestion

A. Algorithm and Complexity *[BPSC- 2021]*

- Definition of Algorithm and characteristics of an algorithm
- Methods of representing algorithms: pseudocode, flowchart, programming language
- Time complexity and space complexity
- Best case, average case and worst case
- Big-O, Big-Omega (Ω), Big-Theta (Θ)
- Complexity analysis of single and nested loops
- Recursion and recursive algorithms
- Divide and Conquer vs Dynamic Programming (Difference, Algo, Sudo Code)
- Greedy Algorithm vs Dynamic Programming (Difference, Algo, Sudo Code)
- Asymptotic growth-rate comparison

B. Linear Data Structures *[BPSC-SAE-2021]*

- Linear vs Non-linear Data Structure
- Array: traversal, insertion, deletion, searching
- Linked List: singly, doubly and circular linked list *[SO IT 2026]*
- Linked List insertion, deletion and reversal
- XOR Linked List: forward and backward traversal
- Stack: PUSH, POP, PEEK/TOP
- Queue: ENQUEUE, DEQUEUE, FRONT/REAR
- Circular Queue and Deque *[OFCIT 2026]*
- Priority Queue: insertion, deletion and updating

- Array vs Linked List
- Stack vs Queue
- Array implementation vs Linked List implementation

C. Searching and Sorting *[Combined SO IT- 2019]*

- Linear Search and Binary Search *[SO IT 2026]*
- Binary Search pseudocode and complexity
- Bubble, Selection and Insertion Sort
- Merge Sort, Quick Sort and Heap Sort
- Stable vs Unstable Sorting
- In-place vs Out-of-place Sorting
- Ascending and descending order sorting
- Array state after each sorting pass
- Best, average and worst case complexities of sorting algorithms
- Choosing a suitable searching/sorting algorithm for a scenario

D. Tree and Heap *[RBL-ANE-2021]*

- Tree terminology: root, leaf, degree, height, depth, level
- Binary Tree and Binary Search Tree (BST)
- Full, complete, perfect and balanced binary tree
- Preorder, Inorder and Postorder traversal
- Tree construction from Inorder and Preorder traversals
- BST search, insertion and deletion
- Min Heap and Max Heap construction
- Heap insertion, deletion and heapify
- Heap Sort
- AVL Tree rotation and balancing basics
- Binary Tree vs BST

E. Stack Applications and Expression *[AP -2026]*

- Infix, Prefix and Postfix expressions *[Officer(IT/ICT)-2019]*
- Infix to Postfix conversion, Infix to Prefix conversion
- Postfix and Prefix expression evaluation, Stack operation. *[BPSC-SAE.-2021]*
- Operator precedence and associativity
- Parentheses matching
- Role of Stack in function calls and recursion
- Array-based vs Linked List-based Stack implementation

F. Graph and Advanced Data Structure *[DBBL-2016]*

- Graph terminology: vertex, edge, degree, path and cycle
- Directed vs Undirected Graph
- Weighted vs Unweighted Graph
- Adjacency Matrix vs Adjacency List

- Breadth-First Search (BFS)
- Depth-First Search (DFS)
- BFS/DFS traversal problems *[AP 2026]*
- Hash Table and collision resolution
- Direct addressing, chaining and open addressing
- Disjoint Set / Union-Find basics
- Abstract Data Type (ADT)
- Dynamic Programming with Fibonacci series
- Memoization vs Tabulation

3. Short Questions

1. What is an algorithm? Write four characteristics of an algorithm.
2. Mention three methods of representing an algorithm with examples.
3. Define Linear and Non-linear Data Structures with two examples of each.
4. What is an Abstract Data Type (ADT)? Give an example.
5. Differentiate between Stack and Queue.
6. What are Overflow and Underflow in a Stack?
7. What is a Circular Queue? Why is it needed?
8. Define Priority Queue. How does it differ from a normal Queue?
9. What is a Linked List? Mention its advantages over an Array.
10. Differentiate between Singly Linked List and Doubly Linked List.
11. What is a Binary Search Tree (BST)? State its ordering property.
12. What is a Heap? Differentiate between Min Heap and Max Heap.
13. Define Stable Sorting and In-place Sorting.
14. What are Best Case, Average Case and Worst Case complexities?
15. What is the difference between BFS and DFS?
16. Define Hashing and Collision Resolution.
17. Differentiate between Memoization and Tabulation.
18. Differentiate between Divide and Conquer and Dynamic Programming.
19. What is the difference between O(1) and O(log n) time complexity?
20. Why is Binary Search unsuitable for a standard unsorted Array?

4. Medium Questions

21. Write the algorithm for inserting an element into a Stack using an Array.
22. Write the steps of ENQUEUE and DEQUEUE operations of a Circular Queue.
23. Write an algorithm to search for an element in a Binary Search Tree.
24. Explain Linear Search and Binary Search with their time complexities.
25. Write Bubble Sort and determine its best-case and worst-case complexities.
26. Explain Selection Sort and Insertion Sort with an example.
27. Explain Merge Sort using the Divide and Conquer approach.
28. Compare Bubble, Insertion, Merge, Quick and Heap Sort by complexity and stability.
29. Explain Infix-to-Postfix conversion using a Stack.
30. Explain how to evaluate a Postfix Expression using a Stack.
31. Construct a Binary Tree from given Preorder and Inorder traversals.
32. Explain Min Heap insertion with an example.

33. Describe BFS traversal of a Graph with an example.
34. Explain Dynamic Programming for generating Fibonacci numbers.
35. Explain backward traversal of a Doubly Linked List.
36. Explain factors affecting time and space complexity.
37. Write a C program to sort five positive integers in ascending order.
38. Compare Array-based and Linked List-based Stack implementations.
39. Explain traversal, searching, insertion, deletion, sorting and merging operations.
40. Explain Hashing and two collision-resolution techniques.

5. Problem-Solving Practice Set

Problem 1 — Searching

An Array contains 220 integers in ascending order. Write pseudocode to find x using a suitable searching algorithm. Determine its worst-case time complexity.

Problem 2 — Stack / Postfix

Evaluate the postfix expression using a Stack and show the Stack contents after each operation: 5 2 3 * + 8 4 / -.

Problem 3 — Tree Construction

Construct a Binary Tree and write its Postorder traversal. Preorder: A B D E C F; Inorder: D B E A C F.

Problem 4 — Min Heap

Construct a Min Heap by inserting 12, 7, 15, 3, 9, 1, 10 in that order. Draw the heap after every insertion.

Problem 5 — Bubble Sort

Sort 7, 3, 5, 2, 4 in ascending order using Bubble Sort. Show each pass and count comparisons and swaps.

Problem 6 — Priority Queue

A Priority Queue contains (A,3), (B,1), (C,2), (D,1). A smaller number means higher priority; FIFO breaks ties. Show deletion order.

Problem 7 — Graph

For an undirected graph with edges A–B, A–C, B–D, B–E, C–F, perform BFS and DFS from A. Visit adjacent vertices alphabetically.

Problem 8 — Complexity

Determine the time complexity and explain: for (int i = 1; i < n; i *= 2) { printf("%d ", i); }

❖ Problem-Solving Answer Key

Problem	Answer / Result
1. Binary Search	Worst-case O(log n). For 220 elements, at most 8 element checks in standard iterative binary search.
2. Postfix	5 2 3 * + 8 4 / - = 5 + 6 - 2 = 9
3. Tree	Root A; left subtree B with children D,E; right subtree C with right child F. Postorder: D E B F C A.
4. Min Heap	Final level order: 1, 3, 7, 12, 9, 15, 10.

5. Bubble Sort	Sorted array: 2, 3, 4, 5, 7. Standard left-to-right pass 1 has 4 swaps; pass 2 has 1 swap; optimized version stops after a no-swap pass.
6. Priority Queue	Deletion order: B, D, C, A.
7. BFS	A, B, C, D, E, F.
7. DFS	A, B, D, E, C, F.
8. Complexity	O(log n), because i doubles in each iteration.

6. Sorting Algorithm Complexity Chart

Algorithm	Best Case	Average Case	Worst Case	Stable?
Bubble Sort (optimized)	O(n)	$O(n^2)$	$O(n^2)$	Yes
Selection Sort	$O(n^2)$	$O(n^2)$	$O(n^2)$	Usually No
Insertion Sort	O(n)	$O(n^2)$	$O(n^2)$	Yes
Merge Sort	O(n log n)	O(n log n)	O(n log n)	Yes
Quick Sort	O(n log n)	O(n log n)	$O(n^2)$	No
Heap Sort	O(n log n)	O(n log n)	O(n log n)	No

Note: Bubble Sort-এর Best Case O(n) হয় যদি implementation-এ কোনো swap না হলে loop বন্ধ করার optimization থাকে। Quick Sort-এর worst case pivot selection-এর ওপর নির্ভর করে।

কোন Sorting Algorithm কখন কার্যকর হতে পারে?

- Nearly sorted data: Insertion Sort কার্যকর হতে পারে.
- Guaranteed O(n log n) comparison-based sorting: Merge Sort বা Heap Sort.
- Quick Sort সাধারণত average case-এ দ্রুত, তবে worst case $O(n^2)$ হতে পারে.
- Bubble Sort সহজে বোঝা যায়, কিন্তু বড় dataset-এর জন্য সাধারণত উপযুক্ত নয়.

7. Searching ও Data Structure Complexity Chart

Operation / Algorithm	Typical Time Complexity
Array index access	O(1)
Linear Search	O(n)
Binary Search (sorted data)	O(log n)
Hash Table Search	Average O(1), Worst O(n)
Stack Push / Pop	O(1)
Queue Enqueue / Dequeue	O(1), suitable implementation
Linked List Search	O(n)
Insertion at known linked-list node	O(1), if pointers are available
BST Search	Average O(log n), Worst O(n)

Balanced BST Search	O(log n)
Heap Insertion	O(log n)
Heap peek min/max	O(1)
Heap extract min/max	O(log n)
BFS / DFS (adjacency list)	O(V + E)

n = number of elements; V = number of vertices; E = number of edges.

8. Revision Checklist

- Time Complexity and Big-O
- Linear Search and Binary Search
- Stack, Queue and Priority Queue
- Linked List and its operations
- Bubble, Insertion, Selection and Merge Sort
- Tree Traversal and Tree Construction
- BST Search, Insertion and Deletion
- Min Heap and Max Heap
- Infix, Prefix, Postfix and Expression Evaluation
- BFS, DFS and Graph Representation
- Recursion and Dynamic Programming
- Hashing and Collision Resolution
- Array, Pointer, Stack and Queue
- Algorithm, Pseudocode and Flowchart

9. Important Questions

1. Describe step-by-step how Binary Search locates a target value in a sorted array. Why does it fail if the array is unsorted? *[AP 2026]*
2. You have two stacks. Explain the logic required to implement a Queue (FIFO) using only these two stacks. *[AP 2026]*
3. Explain the logic of Bubble Sort. Why is it considered inefficient for large datasets compared to Merge Sort? *[AP 2026]*
4. Draw or describe a Flowchart to determine the largest of three distinct numbers (A, B, C). *[AP 2026]*
5. Construct a logical argument explaining why a heuristic (like A* search) might be faster than a blind search (like BFS), even if it doesn't guarantee the absolute perfect path in all cases. *[AP 2026]*
6. In a B+ tree where each node can have a maximum of 4 child pointers, what is the maximum total number of keys that can be stored across all levels of a tree of height 2? *[AP 2026]*
7. Consider the execution of Merge Sort on an unsorted array of n elements. State its exact Time Complexity across Best, Average, and Worst cases. *[AP 2026]*
8. What is Linked list? What is searching? Explain Binary Search. *[SO IT 2026]*
9. Write a real-life example where Depth-First search (DFS) performs more efficiently than Breadth-First Search(BFS). Why is DFS better in that situation? *[DPDC 2025]*
10. What is algorithm? Write the names of three methods of expressing algorithm. *[AP ICT ministry-2021]*
11. You want to search a specific number x in the array where 220 numbers are stored in ascending order. Write down a pseudo code to search x in an optimal search policy. *[APSCL-2021]*

12. Priority queue, updating the queue from the given problem. *[Teletalk-CSE-23]*
13. Difference between Divide and Conquer and Dynamic Programming*[BCC AP-2021]*
14. Running time of data structure search where time complexity is O(1).which algorithm should we use?*[BCC AP-2021]*
15. Queue is an abstract data structure. A queue is open at both its ends. One end is always used to insert data (enqueue) and the other is used to remove data (dequeue). Write the steps of Enqueue Operation of Queue.[officer(SBL-JBL-IT)-2020]
16. What is dynamic programming? Explain with example? Consider an example of the Fibonacci series. The following series is the Fibonacci series: 0 1 1 2 3 5 8 13…… [DMTCL-AE(ICT-23)]
17. There are no well-defined standards for writing algorithms. Efficiency of an algorithm depends on several factors. Similarly, complexity of an algorithm also depends of several factors. Describe the algorithm complexity factors.[officer(SBL-JBL-IT)-2020]
18. Write down some default data types in C.[BCC-project-2021]
19. Write a c program to check the number is EVEN or ODD.[BCC-project-2021]
20. construct a binary tree from the given inorder and preorder sequence.*[RBL-ANE-2021]*
21. Inorder Traversal : { 4, 2, 1, 7, 5, 8, 3, 6 }, Preorder Traversal: { 1, 2, 4, 3, 5, 7, 8, 6 }
22. A doubly linked list is given(Address of next node has XOR operation) Traverse method of the list from last to first[Microcredit Regulatory Authority-AME-2021]
23. Bubble sort algorithm টি লিখুন। এই algorithm টির Time Compexityনির্ণয় করুন। *[BPSC-(M.o.Commercee)-2021]*
24. নিম্নলিখিত Expression টি evaluate করুন।*[BPSC-SAE.-2021]*
    3 2 *2 1 5 3 – 8 4 / * –
25. Stack এর operation গুলো সংক্ষেপে বর্ণনা করুন।*[BPSC-SAE.-2021]*
26. Linear search আলগরিদম কি? এই আলগরিদম এর best case, worst case এবং average case complexity বর্ণনা করুন।*[BPSC-SAE.-2021]*
27. Difference between linear and Nonlinear data structure. *[Pally Sa.-2018]*
28. Describe infix to postfix conversion method. Advantage of stack using link list then array. *[Basic Bank(AM)-2019]*
29. What are the operations performed on a data structure? What is prefix, post fix and infix operation? *[Officer(IT/ICT)-2019]*
30. Draw a Binary min heap that results from inserting 9, 3, 7, 4, 8, 2, 6, 5 in that order into an initially empty heap. Show each insertion step separately. You do not need to show the array representation. Just draw the trees. *[Uttara Bank(AP)-2019]*
31. Describe four types of sorting algorithm with example. *[Combined(SO-IT/ICT)-2019]*
32. Advantage of merge sort, heap sort, Insertion sort. Advantage and complixity of them*[Basic Bank(AM)-2019]*
33. Draw a flowchart to input five positive numbers, and sort them is ascending order. *[Combined 3 bank (AP)-2018]*
34. Write a algorithm to find a node in a binary search tree. *[Pally Sa.-2018]*
35. Difference between stacks and queue with example. *[Engineering college lecturer -2017]*
36. Write down time complexities of different sorting algorithm. *[DBBL-2016]*
37. Explain BFS Traversal of a graph with an example. *[DBBL-2016]*